

José Federico Castro Tramontina

Software and Embedded Systems Developer | Cybersecurity Specialist

Applied Cryptography · Artificial Intelligence · Linux · Python · C/C++

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 Buenos Aires | Mendoza, Argentina

Open to remote, hybrid, and on-site roles · Willing to relocate

PROFESSIONAL PROFILE

IT professional specializing in software development, embedded systems, applied cryptography, and cybersecurity. Experienced in designing and implementing real-time client-server solutions, process automation, systems administration, and industrial cybersecurity. Argentine Army officer with experience in team leadership, communications, and information security. Able to integrate software, hardware, networking, and security into real-world technology solutions.

TECHNICAL SKILLS

Programming languages: Python, C, C++, JavaScript, Bash, SQL, HTML5, CSS3.

Embedded systems and IoT: Arduino, ESP32, STM32, Raspberry Pi, communications, hardware-software integration, and KiCad.

Cybersecurity: Cryptography, post-quantum cryptography, OT/ICS security, ISO 27001, Modbus TCP/IP, Wireshark, Nmap, and tcpdump.

Scientific computing and local AI: SageMath, Jupyter, MATLAB/GNU Octave, llama.cpp, LM Studio, and ComfyUI.

Backend and data: FastAPI, WebSockets, SQLite, APIs, client-server architecture.

Industrial automation: PLCs, Ladder Logic, Structured Text, TIA Portal, EcoStruxure.

Systems and DevOps: GNU/Linux, Docker, Git, GitHub, OpenSSH, Samba, Cron, and VirtualBox.

PROFESSIONAL EXPERIENCE

Software and Embedded Systems Developer — Freelance

May 2026 – Present

Real-time queue management system currently running in production at one Farmacia Vital Olivos branch. Client-server architecture comprising three applications: a customer touchscreen kiosk, an operator console, and a multimedia queue display. Developed with Python, FastAPI, WebSockets, SQLite, HTML5, CSS3, and JavaScript.

Cybersecurity Consultant — Freelance

November 2024 – Present

Cybersecurity advisory services for genomIT focused on regulatory compliance, personal data protection, and strengthening information security policies and controls aligned with ISO 27001.

IT Officer | Systems Administrator — Argentine Army

June 2014 – Present

Systems administration, automation with Python, Bash, and Google Apps Script, Moodle platform maintenance, and configuration of computing and radio communications equipment. Responsible for communications and information security. Served as Section Leader, Echelon Leader, Communications Officer, and leader of teams of up to 40 personnel.

KEY PROJECTS

Post-quantum key exchange protocol: design, implementation, and cryptanalysis of a protocol based on Cayley-Dickson algebras, including evaluation through cryptographic attacks.

Operational technology cybersecurity: industrial network design, Modbus TCP/IP vulnerability analysis, PLC programming, event monitoring, and process automation for OT environments.

Secure embedded systems network: design and implementation of an embedded device architecture protected through post-quantum cryptography.

IFPLab: autonomous local AI research agent for integer factorization, featuring automatic hypothesis generation, SageMath-based computational experimentation, and iterative closed-loop feedback.

EDUCATION

PhD Candidate in Computer Science — Universidad Abierta Interamericana. Dissertation: *Design, Development, and Cryptanalysis of a Key Exchange Protocol Based on Cayley-Dickson Algebras*.

Master's Candidate in Data Networks — Universidad Tecnológica Nacional, Facultad Regional Mendoza. Thesis: *Design and Implementation of a Secure Embedded Systems Network Using Post-Quantum Cryptography*.

Master's Degree in Cyberdefense — Facultad de Ingeniería del Ejército.

Postgraduate Specialization in Cryptography and Telecommunications Security — Facultad de Ingeniería del Ejército.

Bachelor's Degree in Operational Leadership and Management — Facultad del Ejército, Colegio Militar de la Nación.

Electronic Engineering — Facultad de Ingeniería del Ejército. In progress · Full-time scholarship student.

TEACHING AND RESEARCH

University lecturer in Cryptography, Information Security, Discrete Mathematics, and Physics. Researcher in applied cryptography, industrial cybersecurity, and embedded systems. Author and co-author of scientific publications, speaker at national conferences, and master's thesis committee member.

Languages: Spanish: Native · English: B2 (CEFR)